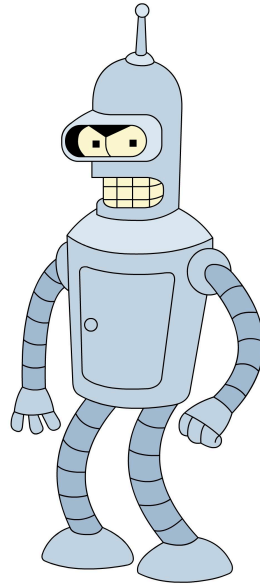


Reinforcement Learning

HSE, winter - spring 2025

Lecture 5: Advanced Policy-Based



Sergei Laktionov
slaktionov@hse.ru
[LinkedIn](#)

Recap: Policy Gradient

$$\nabla J(\theta) = \mathbb{E}_{p_{\theta}(\tau)} \left[\sum_{t=0}^T \nabla \log \pi_{\theta}(a_t | s_t) \Psi_t \right],$$

$$A = Q(s_t, a_t) - V(s_t) \\ r + \gamma V(s') - V(s)$$

where Ψ_t may be one of the following:

- $\sum_{t=0}^T \gamma^t R_t$: total reward of the trajectory

- $\sum_{k=t}^T \gamma^{k-t} R_k$: reward following action a_t

- $Q^{\pi}(s_t, a_t)$: action value function

- $Q^{\pi}(s_t, a_t) - \underline{b(s_t)}$: baseline version of previous formula.

- $A^{\pi}(s_t, a_t)$: advantage function

- $\sum_{k=0}^{N-1} \gamma^k r_{t+k} + \gamma^N V^{\phi}(s_{t+N}) - V^{\phi}(s_t)$:

TD(N) residual

Recap: A2C

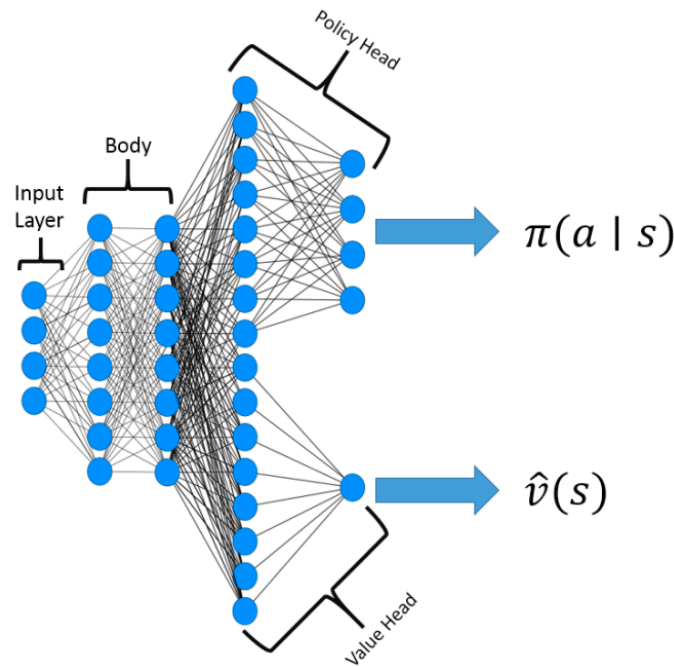
- Generate trajectories $\{\tau_i\}$ following $\pi_\theta(a | s)$ in parallel
- Policy improvement:

- $A^\phi(s_{i,t}, a_{i,t}) = r_{i,t} + \gamma(1 - done_{i,t})V(s_{i,t+1}) - V(s_{i,t})$
- Estimate gradient and make gradient ascent step:

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \left[\sum_{t=0}^T \nabla \log \pi_\theta(a_{i,t} | s_{i,t}) \underline{A^\phi(s_{i,t}, a_{i,t})} \right]$$

- Policy evaluation:
- Estimate gradient and make gradient descent step:

$$\nabla_\phi L(\phi) \approx \frac{1}{N} \sum_{i=1}^N \left[\sum_{t=0}^T \nabla_\phi (r_{i,t} + \gamma \underbrace{V_\phi(s_{i,t+1})}_{\text{Frozen parameters}} - V_\phi(s_{i,t}))^2 \right]$$



Source

Policy Gradient

$$\nabla J(\theta) = \mathbb{E}_{p_{\theta}(\tau)} \left[\sum_{t=0}^T \nabla \log \pi_{\theta}(a_t | s_t) A^{\pi}(s_t, a_t) \right]$$

The choice $A^{\pi}(s_t, a_t)$ yields almost the lowest possible variance, though in practice, the advantage function is not known and must be estimated.

$$b(s) = c \cdot V(s)$$

Advantage Estimator

- Let V be an approximate value function

- $\delta_t := r_t + \gamma V(s_{t+1}) - V(s_t)$

- $A_t^{(1)} = r_t + \gamma V(s_{t+1}) - V(s_t) = \delta_t$

$$\gamma \cdot [r_{t+1} + \gamma V(s_{t+2}) - V(s_{t+1})]$$

- $$A_t^{(2)} = r_t + \gamma r_{t+1} + \gamma^2 V(s_{t+2}) - V(s_t) =$$

$$= r_t + \underbrace{\gamma V(s_{t+1}) - V(s_t)}_{\delta_t} + \underbrace{\gamma r_{t+1} + \gamma^2 V(s_{t+2}) - \gamma V(s_{t+1})}_{\gamma \delta_{t+1}} = \delta_t + \gamma \delta_{t+1}$$

...

- $$A_t^{(N)} = r_t + \gamma r_{t+1} + \dots + \gamma^{N-1} r_{t+N-1} + \gamma^N V(s_{t+N}) - V(s_t) = \sum_{k=0}^{N-1} \gamma^k \delta_{t+k}$$

- $$A_t^{(\infty)} = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k}$$

Advantage Estimator

- Let V be an approximate value function

- $\delta_t := r_t + \gamma V(s_{t+1}) - V(s_t)$

- $A_t^{(1)} = r_t + \gamma V(s_{t+1}) - V(s_t) = \delta_t$

- $$A_t^{(2)} = r_t + \gamma r_{t+1} + \gamma^2 V(s_{t+2}) - V(s_t) =$$

$$= r_t + \gamma V(s_{t+1}) - V(s_t) + \gamma r_{t+1} + \gamma^2 V(s_{t+2}) - \gamma V(s_{t+1}) = \delta_t + \gamma \delta_{t+1}$$

Low variance

High bias

...

- $$A_t^{(N)} = r_t + \gamma r_{t+1} + \dots + \gamma^{N-1} r_{t+N-1} + \gamma^N V(s_{t+N}) - V(s_t) = \sum_{k=0}^{N-1} \gamma^k \delta_{t+k}$$

High variance

Low bias

- $$A_t^{(\infty)} = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k}$$

Generalised Advantage Estimator

$$\bullet A_t^{(N)} = \sum_{k=0}^{N-1} \gamma^k \delta_{t+k}$$

$\lambda \in [0; 1] \Rightarrow \text{ensemble}$

- GAE is defined as the exponentially-weighted average of these N-step estimators, where λ is the hyperparameter that controls bias-variance trade-off:

$$A_t^{GAE(\gamma, \lambda)} = (1 - \lambda)(A_t^{(1)} + \lambda A_t^{(2)} + \dots) =$$

$$= (1 - \lambda) \cdot (\underbrace{s_t}_{\text{green}} + \lambda (\underbrace{s_t}_{\text{green}} + \underbrace{\gamma \delta_{t+1}}_{\text{teal}}) + \lambda^2 (\underbrace{s_t}_{\text{green}} + \underbrace{\gamma \delta_{t+1}}_{\text{teal}} + \underbrace{\gamma^2 \delta_{t+2}}_{\text{teal}}) + \dots)$$

$$= (1 - \lambda) \cdot \left[\frac{\underbrace{s_t}_{\text{green}}}{\underbrace{(1 - \lambda)}} + \frac{\lambda \underbrace{\gamma \delta_{t+1}}_{\text{teal}}}{\underbrace{(1 - \lambda)}} + \frac{\lambda^2 \gamma \delta_{t+2}}{\underbrace{(1 - \lambda)}} + \dots \right] = \sum_{k=0}^{\infty} (\gamma \lambda)^k \delta_{t+k}$$

Generalised Advantage Estimator

- $A_t^{(N)} = \sum_{k=0}^{N-1} \gamma^k \delta_{t+k}$
- GAE is defined as the exponentially-weighted average of these N-step estimators, where λ is the hyperparameter that controls bias-variance trade-off:

$$A_t^{GAE(\gamma, \lambda)} = (1 - \lambda)(A_t^{(1)} + \lambda A_t^{(2)} + \dots) = \sum_{k=0}^{\infty} (\gamma \lambda)^k \delta_{t+k}$$

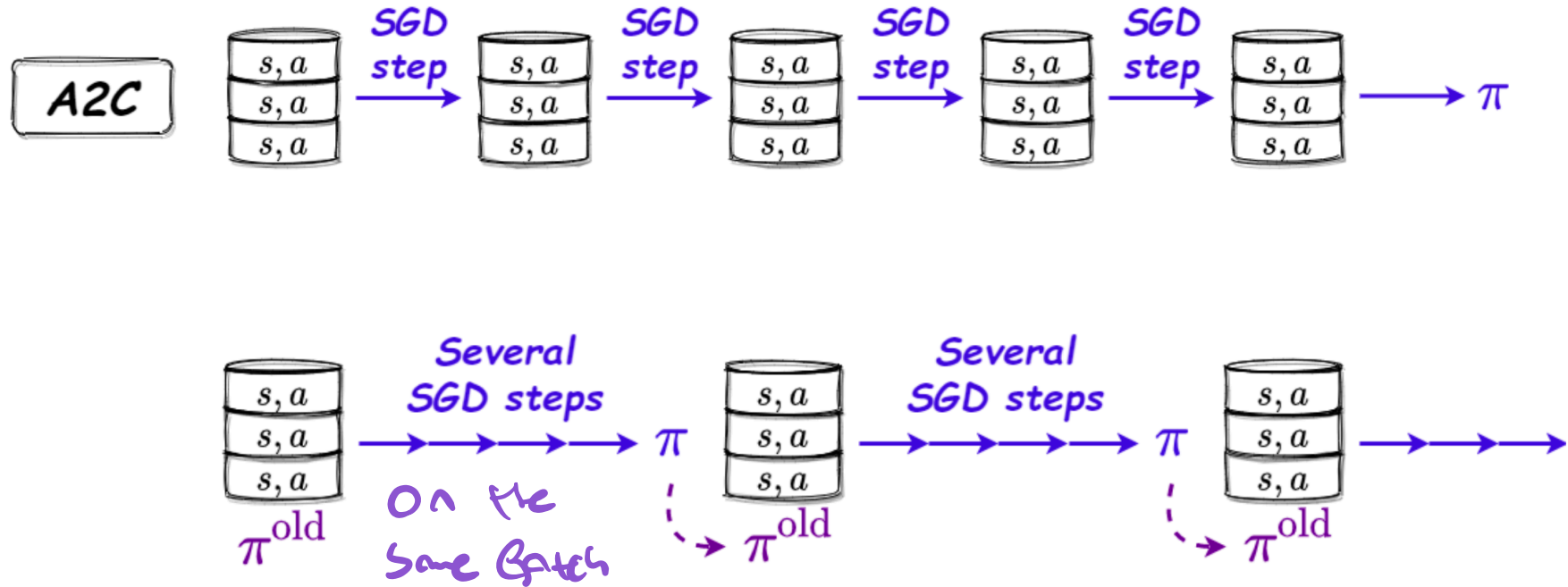
What is the benefit?

As our critic is not ideal, we want to use as much collected information as possible in order to decrease the variance

Sample Efficiency



Sample Efficiency



Sample Efficiency



A2C



Source

Reward Shaping

Theorem [Reward Shaping]:

Imagine we are given a function $\Phi : \mathcal{S} \rightarrow \mathbb{R}$, which we call a **potential**. We require the function to be bounded and to satisfy $\Phi(s^{terminal}) = 0$.

Then using the following new reward function corresponds to the equivalent task:

$$r^{new}(s, a, s') := r^{initial}(s, a) + \gamma\Phi(s') - \Phi(s).$$

Proof:

Imagine we are given some trajectory τ . Then $G_0 = \sum_{\tau} \gamma^t r_t$.

For a new reward we have

$$\tilde{G}_0 = \sum_{\tau} \gamma^t r_t + \sum_{\tau} [\gamma^{t+1}\Phi(s_{t+1}) - \gamma^t\Phi(s_t)] = \{\text{telescoping sum}\} = \sum_{\tau} \gamma^t r_t + \gamma^T\Phi(s_T) - \Phi(s_0).$$

Then $\tilde{J}(\pi) \equiv J(\pi)$, as $\Phi(s_0)$ is independent of the policy and $\lim_{t \rightarrow +\infty} \gamma^t \Phi(s_t) = 0$ or $\Phi(s_T) = 0$.

Relative Performance Improvement

Theorem [Relative Performance Improvement]:

For any two valid policies π_1, π_2 it is true that

$$\underline{V^{\pi_2}(s) - V^{\pi_1}(s)} = \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \sum_{t \geq 0} \gamma^t \underline{A^{\pi_1}(s_t, a_t)}$$

Proof:

$$V^{\pi_2}(s) - V^{\pi_1}(s) = \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \sum_{t \geq 0} \gamma^t r_t - V^{\pi_1}(s) = \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \left[\sum_{t \geq 0} \gamma^t r_t - \underline{V^{\pi_1}(s_0)} \right]$$

Telescoping sum $\longrightarrow$ $= \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \left[\sum_{t \geq 0} \gamma^t r_t + \sum_{t \geq 0} (\gamma^{t+1} V^{\pi_1}(s_{t+1}) - \gamma^t V^{\pi_1}(s_t)) \right]$

Rearrange terms $\longrightarrow$ $= \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \sum_{t \geq 0} \gamma^t [r_t + \gamma V^{\pi_1}(s_{t+1}) - V^{\pi_1}(s_t)]$

$\mathbb{E}_x$ is idempotent $\longrightarrow$ $= \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \sum_{t \geq 0} \gamma^t [r_t + \gamma \mathbb{E}_{s_{t+1}} V^{\pi_1}(s_{t+1}) - V^{\pi_1}(s_t)]$

By definition of Q^{π_1} $\longrightarrow$ $= \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \sum_{t \geq 0} \gamma^t [Q^{\pi_1}(s_t, a_t) - V^{\pi_1}(s_t)] = \mathbb{E}_{\tau \sim \pi_2 | s_0=s} \sum_{t \geq 0} \gamma^t A^{\pi_1}(s_t, a_t)$

$\nwarrow$ Reward Shaping
 $Q(s) \equiv V(s)$

Policy Improvement

On each step we would like to have a positive difference $J(\pi) - J(\pi_{old})$

$$\begin{aligned} \text{Note that } \underline{J(\pi) - J(\pi_{old})} &= J(\pi) - \mathbb{E}_{\tau \sim \pi_{old}}[V^{\pi_{old}}(s_0)] = \\ &= \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] + \mathbb{E}_{\tau \sim \pi_{old}} \left[\sum_{t=0}^{\infty} \gamma V^{\pi_{old}}(s_{t+1}) - \sum_{t=0}^{\infty} \gamma V^{\pi_{old}}(s_t) \right] = \\ &= \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t (r_t + \gamma V^{\pi_{old}}(s_{t+1}) - V^{\pi_{old}}(s_t)) \right] = \\ &= \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t \underline{A^{\pi_{old}}(s_t, a_t)} \right] \end{aligned}$$

$$\pi_1 = \pi_{old}$$

$$\pi_2 = \pi$$

Policy Improvement

$$J(\pi) - J(\pi_{old}) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_{old}}(s_t, a_t) \right] \geq 0$$

1. Let's define $\pi(s) = \operatorname{argmax}_a A^{\pi_{old}}(s, a)$. Then $\sum_{t=0}^{\infty} \gamma^t A^{\pi_{old}}(s_t, a_t) \geq 0$
and we guarantee policy improvement on each step.

2. It's enough to have $\mathbb{E}_{a \sim \pi(\cdot|s)} A^{\pi_{old}}(s, a) \geq 0$

Alternative Form

$$J(\pi) - J(\pi_{old}) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_{old}}(s_t, a_t) \right]$$

Lemma:

- Define discounted state-visitation distribution:

$$d_{\pi}(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s \mid \pi)$$

- Then for all $f(s, a)$: $\mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \right] = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi}} \mathbb{E}_{a \sim \pi(\cdot | s)} [f(s, a)]$

Alternative Form

$$J(\pi) - J(\pi_{old}) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_{old}}(s_t, a_t) \right]$$

Lemma:

- Define discounted state-visitation distribution:

$$d_{\pi}(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s \mid \pi)$$

- Then for all $f(s, a)$: $\mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \right] = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi}} \mathbb{E}_{a \sim \pi(\cdot | s)} [f(s, a)]$

Alternative Form

$$J(\pi) - J(\pi_{old}) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi_{old}}(s_t, a_t) \right]$$

Lemma:

- Define discounted state-visitation distribution:

$$d_{\pi}(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \mathbb{P}(s_t = s \mid \pi)$$

- Then for all $f(s, a)$: $\mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t f(s_t, a_t) \right] = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi}} \mathbb{E}_{a \sim \pi(\cdot | s)} [f(s, a)]$

“Where the agent spends his time during the episode under the policy π ”

It can be viewed as a heat map over states that mixes environment stochasticity and policy stochasticity

Theorem 5.7 in RL Textbook in Russian

Alternative Form π_{old}

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \left[\underbrace{\mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [A^{\pi_{old}}(s, a)]}_{\textcircled{1}} \right]$$

$$\rightarrow \mathbb{E}_{a \sim \pi_\theta(\cdot|s)} [A^{\pi_{old}}(s, a)] = \int A^{\pi_{old}}(s, a) \pi_\theta(a|s) da$$

$$\begin{aligned} &= \int A^{\pi_{old}}(s, a) \frac{\pi_\theta(a|s)}{\pi_{old}(a|s)} \cdot \pi_{old}(a|s) da \\ &= \mathbb{E}_{a \sim \pi_{old}(\cdot|s)} \left[\frac{\pi_\theta(a|s)}{\pi_{old}(a|s)} \cdot A^{\pi_{old}}(s, a) \right] \end{aligned}$$

Alternative Form

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot | s)} [A^{\pi_{old}}(s, a)]$$

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} \left[\frac{\pi_\theta(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right]$$

Alternative Form

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot | s)} [A^{\pi_{old}}(s, a)]$$

$$\frac{0.95}{0.01}$$

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} \left[\frac{\pi_\theta(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right]$$

absolutely
continuous

For importance-weighting we

require π_θ w.r.t. π_{old} , i.e.,

$$\pi_{old}(a | s) = 0 \implies \pi_\theta(a | s) = 0$$

(if we use the Softmax / Gaussian, this
is automatically satisfied)

$$\mathcal{N}(\mu_\theta, \sigma_\theta^2)$$

Alternative Form

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \mathbb{E}_{a \sim \pi_\theta(\cdot | s)} [A^{\pi_{old}}(s, a)]$$

$$J(\pi_\theta) - J(\pi_{old}) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_\theta}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} \left[\frac{\pi_\theta(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right]$$

Define a surrogate objective:

$$L_{\pi_{old}}(\theta) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_{old}}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} \left[\frac{\pi_\theta(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right]$$

Alternative Form

A surrogate objective:

$$\begin{aligned} L_{\pi_{old}}(\theta) &= \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_{\pi_{old}}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} \left[\frac{\pi_{\theta}(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right] \\ &= \mathbb{E}_{\tau \sim \pi_{old}} \sum_{t \geq 0} \gamma^t \left[\frac{\pi_{\theta}(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right] \end{aligned}$$

Proof:

See Theorem 57 (page 127) in RL Textbook

Optimisation in Policy Space

$$\text{Let } D_{KL}(\pi_{old} || \pi_{\theta}) = \mathbb{E}_{s \sim d_{\pi_{old}}} [D_{KL}(\pi_{old}(\cdot | s) || \pi_{\theta}(\cdot | s))]$$

*

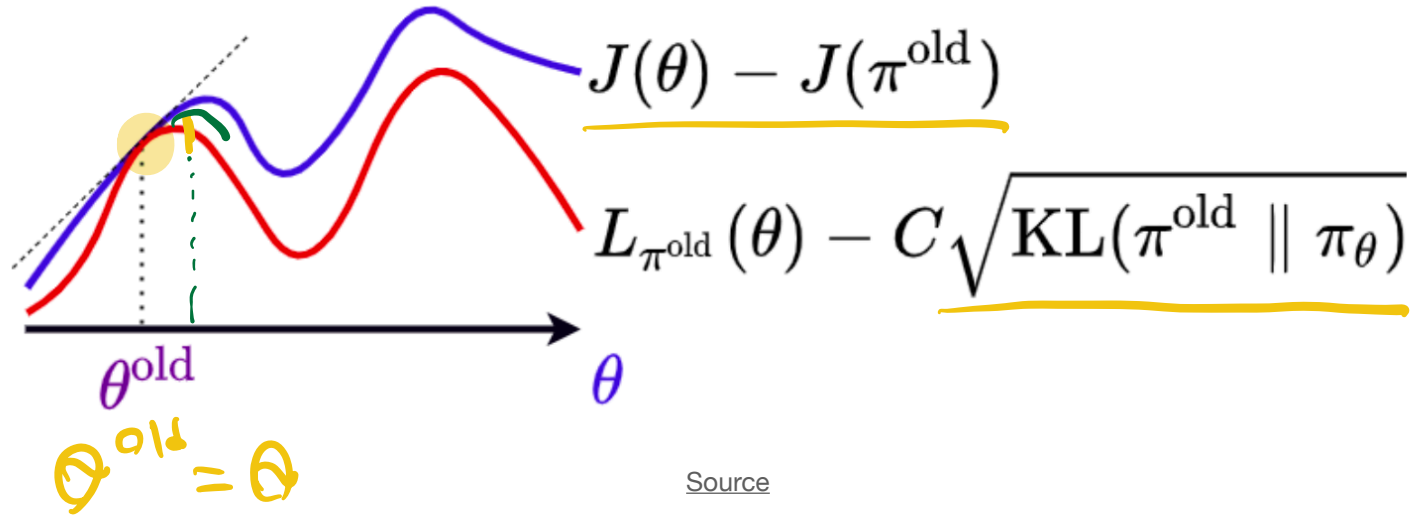
Improvement Lower Bound:

$$\underline{J(\pi_{\theta}) - J(\pi_{old})} \geq L_{\pi_{old}}(\theta) - C \sqrt{D_{KL}(\pi_{old} || \pi_{\theta})},$$

$$\text{Where } C = \frac{\sqrt{2}\gamma}{(1-\gamma)^2} \max_{s,a} |A^{\pi_{old}}(s, a)|$$

$$D_{KL} = \int_{-\infty}^{+\infty} \pi_{old} \ln \frac{\pi_{old}}{\pi_{\theta}} ds$$

Optimisation in Policy Space



Source

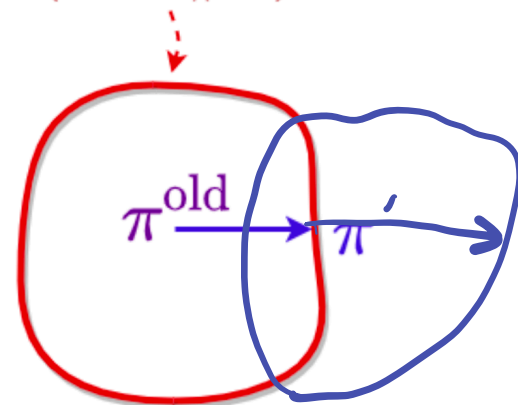
Trust Region Policy Optimisation (TRPO)

$$L_{\pi_{old}}(\theta) \rightarrow \max_{\theta}$$

$$\text{s.t. } D_{KL}(\pi_{old} || \pi_{\theta}) \leq \delta$$

Source

$$\text{KL}(\pi^{\text{old}} || \pi) \leq \delta$$



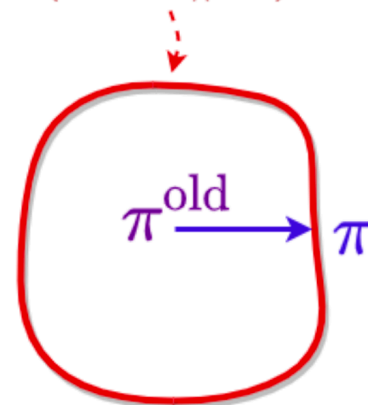
Trust Region Policy Optimisation (TRPO)

Source

$$L_{\pi_{old}}(\theta) \rightarrow \max_{\theta}$$

$$\text{s.t. } D_{KL}(\pi_{old} || \pi_{\theta}) \leq \delta$$

$$\text{KL}(\pi^{old} || \pi) \leq \delta$$



$$L_{\pi_{old}}(\theta) \approx g^T(\theta - \theta_{old}), \text{ where } g = \nabla_{\theta} L_{\pi_{old}}(\theta) |_{\theta_{old}}$$

$$D_{KL}(\pi_{\theta_{old}} || \pi_{\theta}) \approx \frac{1}{2}(\theta - \theta_{old})^T K(\theta - \theta_{old}), \text{ where } K = \nabla_{\theta}^2 D_{KL}(\pi_{old} || \pi_{\theta}) |_{\theta_{old}}$$

$$\begin{matrix} \approx 0 + 0 \\ \theta = \theta_{old} \end{matrix}$$

$$\mathbb{E}[-\nabla \log \pi] = 0$$

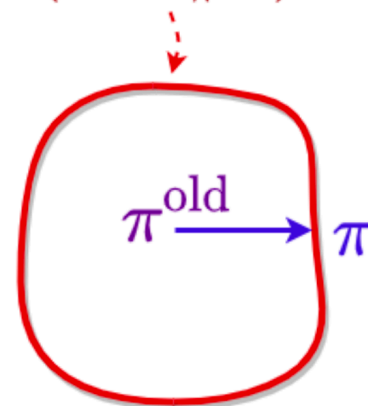
Trust Region Policy Optimisation (TRPO)

Source

$$L_{\pi_{old}}(\theta) \rightarrow \max_{\theta}$$

$$\text{s.t. } D_{KL}(\pi_{old} || \pi_{\theta}) \leq \delta$$

$$KL(\pi^{old} || \pi) \leq \delta$$



$$L_{\pi_{old}}(\theta) \approx g^T(\theta - \theta_{old}), \text{ where } g = \nabla_{\theta} L_{\pi_{old}}(\theta) |_{\theta_{old}}$$

$$D_{KL}(\pi_{\theta_{old}} || \pi_{\theta}) \approx \frac{1}{2}(\theta - \theta_{old})^T K(\theta - \theta_{old}), \text{ where } K = \nabla_{\theta}^2 D_{KL}(\pi_{old} || \pi_{\theta}) |_{\theta_{old}}$$

New task:

$$g^T(\theta - \theta_{old}) \rightarrow \max_{\theta}$$
$$\text{s.t. } \frac{1}{2}(\theta - \theta_{old})^T K(\theta - \theta_{old}) \leq \delta$$

View GD Step as Restricted Search

Standard GA:

$$\begin{aligned} & \underbrace{g^T(\theta - \theta_{old})}_{\theta} \rightarrow \max_{\theta} \\ \text{s.t. } & \underbrace{||\theta - \theta_{old}||_2^2}_{\theta} = \underbrace{(\theta - \theta_{old})^T (\theta - \theta_{old})}_{\theta} \leq \beta \end{aligned}$$

Then optimal step is $\theta_{new} = \theta_{old} + \frac{\sqrt{\beta}}{||g||_2} g$

TRPO New task:

$$\begin{aligned} & \underbrace{g^T(\theta - \theta_{old})}_{\theta} \rightarrow \max_{\theta} \\ \text{s.t. } & \frac{1}{2}(\theta - \theta_{old})^T \underbrace{K}_{\theta} (\theta - \theta_{old}) \leq \delta \end{aligned}$$

View GD Step as Restricted Search

Standard GA:

$$\begin{aligned} g^T(\theta - \theta_{old}) &\rightarrow \max_{\theta} \\ \text{s.t. } ||\theta - \theta_{old}||_2^2 &= (\theta - \theta_{old})^T(\theta - \theta_{old}) \leq \beta \end{aligned}$$

Then optimal step is $\theta_{new} = \theta_{old} + \frac{\sqrt{\beta}}{||g||_2} g$

TRPO New task:

$$\begin{aligned} g^T(\theta - \theta_{old}) &\rightarrow \max_{\theta} \\ \text{s.t. } \frac{1}{2}(\theta - \theta_{old})^T K(\theta - \theta_{old}) &\leq \delta \end{aligned}$$

Intuition:

TRPO update step can be viewed as the same policy gradient (with advantage) update, **but with the constraint in policy space instead of parameter space!**

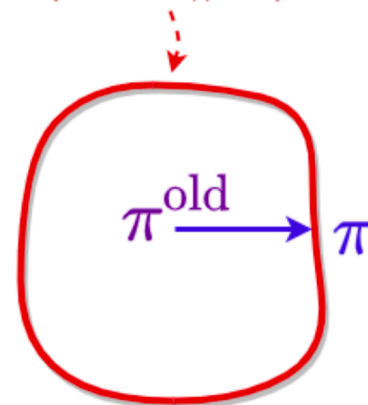
Trust Region Policy Optimisation (TRPO)

Source

$$L_{\pi_{old}}(\theta) \rightarrow \max_{\theta}$$

$$\text{s.t. } D_{KL}(\pi_{old} || \pi_{\theta}) \leq \delta$$

$$KL(\pi^{old} || \pi) \leq \delta$$



$$L_{\pi_{old}}(\theta) \approx g^T(\theta - \theta_{old}), \text{ where } g = \nabla_{\theta} L_{\pi_{old}}(\theta) |_{\theta_{old}}$$

$$D_{KL}(\pi_{\theta_{old}} || \pi_{\theta}) \approx \frac{1}{2}(\theta - \theta_{old})^T K(\theta - \theta_{old}), \text{ where } K = \nabla_{\theta}^2 D_{KL}(\pi_{old} || \pi_{\theta}) |_{\theta_{old}}$$

Optimal solution
for the new task:

$$\theta = \theta_{old} + \alpha \underline{K^{-1}} g, \text{ where } \alpha = \sqrt{\frac{2\delta}{g^T \underline{K^{-1}} g}}$$

$$K \in \mathbb{R}^{|\theta| \times |\theta|}, \underline{K^{-1}} \text{ computation takes } O(|\theta|^3)$$

Conjugate Gradient Method

K is a symmetric, positive-definite matrix

In order to find $K^{-1}g$ we can solve system $Ks = g$ iteratively.

For each iteration $i \leq N_{steps}$ has complexity $O(|\theta|^2)$ and N_{steps} is a hyperparameter that balances precision and speed

Conjugate Gradient Method

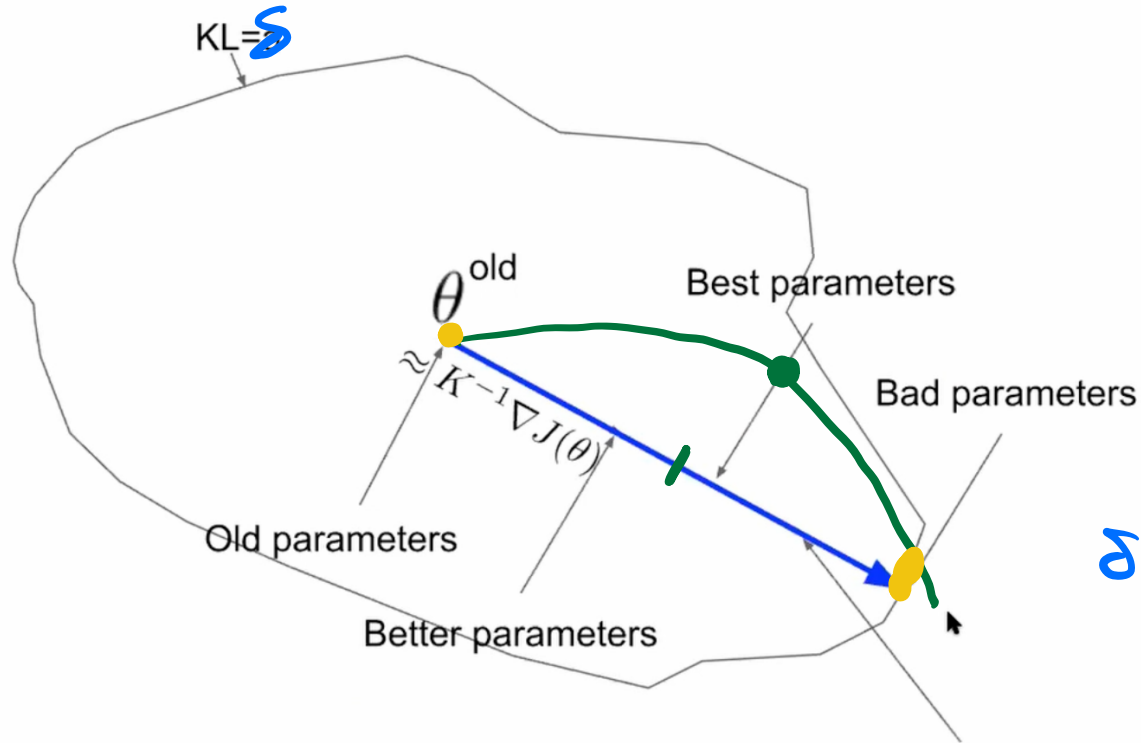
K is a symmetric, positive-definite matrix

In order to find $K^{-1}g$ we can solve system $Ks = g$ iteratively.

For each iteration $i \leq N_{steps}$ has complexity $O(|\theta|^2)$ and N_{steps} is a hyperparameter that balances precision and speed

Another stream of approaches is applying a low-rank approximation of K
— see for example [this paper](#)

Visualisation



We want to compute loss function here!

Source

TRPO Algorithm

Repeat until convergence:

1. Collect trajectories following current policy $\pi_{\theta_{old}}$

2. Compute $g = \nabla_{\theta} \frac{1}{N} \sum_{i=1}^N \frac{\pi_{\theta}(a_i | s_i)}{\pi_{\theta_{old}}(a_i | s_i)} \underbrace{A^{\pi_{\theta_{old}}(s_i, a_i)}}_{\text{handwritten underline}}$

3. Compute $K = \nabla_{\theta}^2 \frac{1}{N} \sum_{i=1}^N D_{KL}(\pi_{\theta_{old}}(\cdot | s_i) || \pi_{\theta}(\cdot | s_i))$

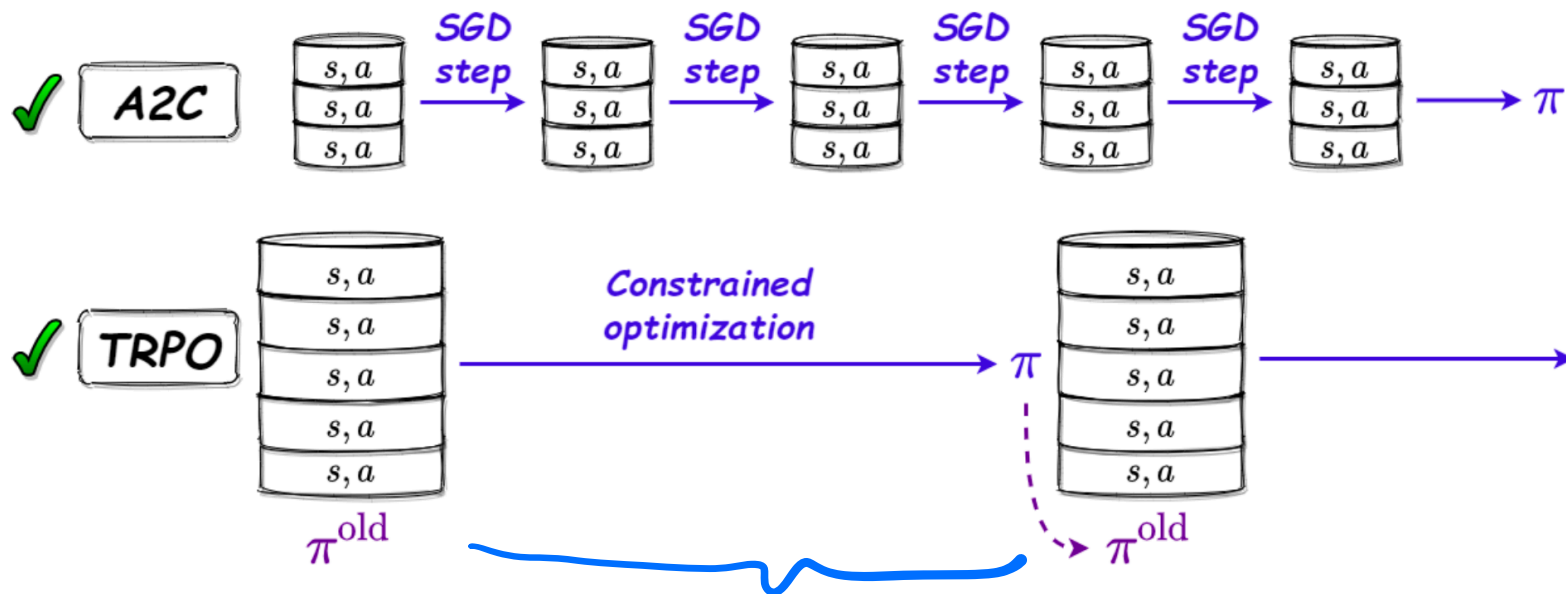
$\mathcal{N}(\mu_{\theta}, \sigma_{\theta}^2)$

4. Find optimal direction via Conjugate Gradients Method (find $s = K^{-1}g$)

5. Do linear search in optimal direction checking the KL constraint and

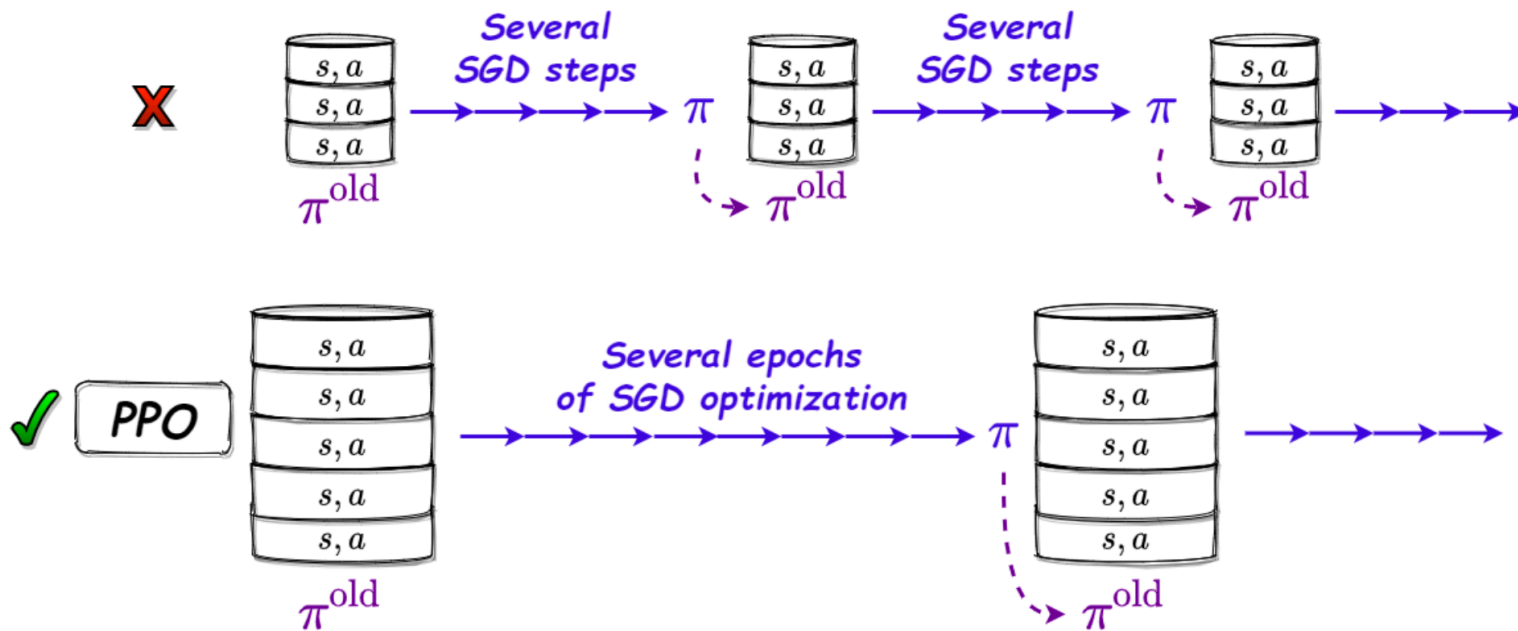
objective value for each new parameter: $\theta_j = \theta_{old} + \underbrace{\alpha_j \sqrt{\frac{2\delta}{g^T s}} s}_{\text{handwritten underline}}$

Comparison



Source

Beyond the Second-order Optimisation



Source

Problem Statement

$$L_{\pi_{old}}(\theta) = \mathbb{E}_{s \sim d_{\pi_{old}}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} \left[\frac{\pi_{\theta}(a | s)}{\pi_{old}(a | s)} A^{\pi_{old}}(s, a) \right]$$

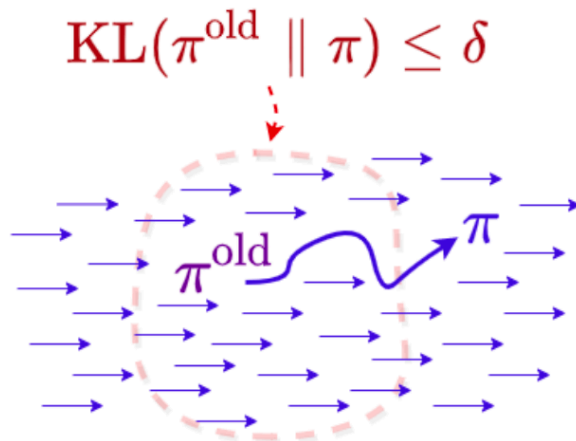
Constrained problem

$$L_{\pi_{old}}(\theta) \rightarrow \max_{\theta}$$

$$\text{s.t. } D_{KL}(\pi_{old} || \pi_{\theta}) \leq \delta$$

Unconstrained problem

$$L_{\pi_{old}}(\theta) - \beta D_{KL}(\pi_{old} || \pi_{\theta}) \rightarrow \max_{\theta}$$



PPO Objective

$$r(\theta) = \frac{\pi_{\theta}(a | s)}{\pi_{\theta_{old}}(a | s)}$$

$$r^{CLIP}(\theta) = \text{clip}\left(\frac{\pi_{\theta}(a | s)}{\pi_{\theta_{old}}(a | s)}, 1 - \epsilon, 1 + \epsilon\right)$$

KL (arrow from $r^{CLIP}(\theta)$ to $\frac{\pi_{\theta}(a | s)}{\pi_{\theta_{old}}(a | s)}$)

0.8 (under $1 - \epsilon$) 1.2 (under $1 + \epsilon$)

TRPO

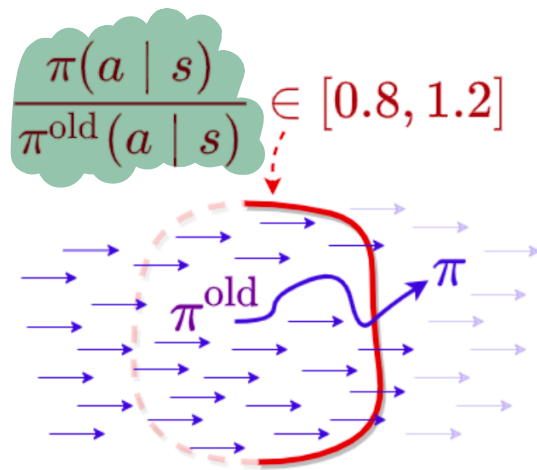
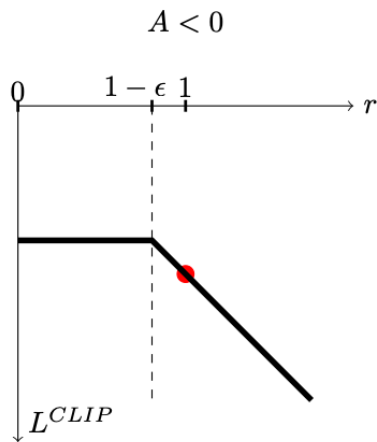
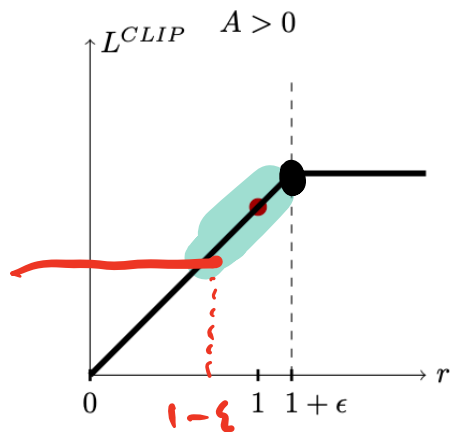


$$L_{\pi_{old}}(\theta) = \mathbb{E}_{s \sim d_{\pi_{old}}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} [r(\theta) A^{\pi_{old}}(s, a)]$$

$$\text{PPO} \rightarrow L_{\pi_{old}}^{CLIP}(\theta) = \mathbb{E}_{s \sim d_{\pi_{old}}} \mathbb{E}_{a \sim \pi_{old}(\cdot | s)} [\min(r(\theta) A^{\pi_{old}}(s, a), r^{CLIP}(\theta) A^{\pi_{old}}(s, a))]$$

PPO Gradient

$$\min(r(\theta)A^{\pi_{old}(s,a)}, r^{CLIP}(\theta)A^{\pi_{old}(s,a)})$$



Source

$A > 0$

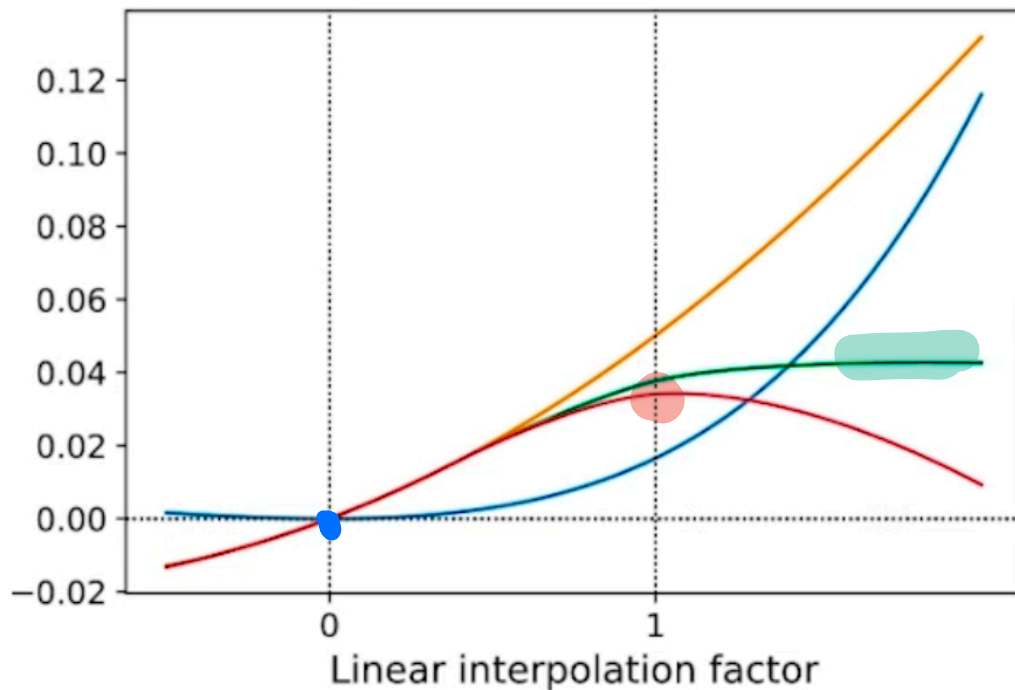
$r(\theta) > 1 + \epsilon \Rightarrow \min(r(\theta)A^{\pi_{old}(s,a)}, r^{CLIP}(\theta)A^{\pi_{old}(s,a)}) \Rightarrow \nabla L = 0$

$r(\theta) \leq 1 + \epsilon \Rightarrow \min(r(\theta)A^{\pi_{old}(s,a)}, r^{CLIP}(\theta)A^{\pi_{old}(s,a)}) \Rightarrow$

$\therefore$ constant

Visualisation

$$\Rightarrow \nabla \pi(\theta) A^{\pi_{\theta}}(s, a) \\ = \frac{\nabla \pi_{\theta}}{\pi_{\theta}} A^{\pi_{\theta}}(s, a)$$

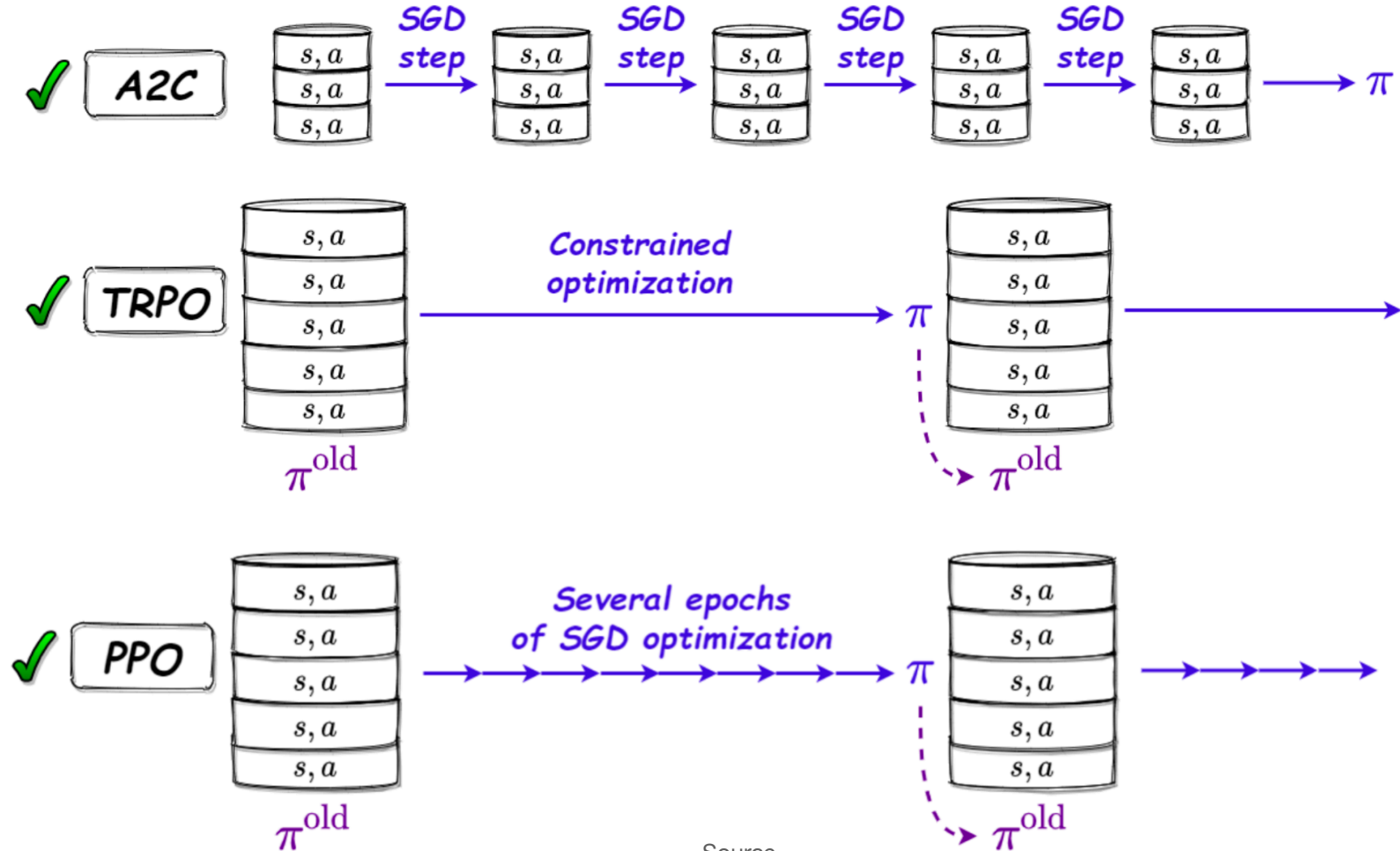


- $\hat{E}_t[KL_t]$
- $L^{CPI} = \hat{E}_t[r_t A_t]$
- $\hat{E}_t[\text{clip}(r_t, 1 - \varepsilon, 1 + \varepsilon) A_t]$
- $L^{CLIP} = \hat{E}_t[\min(r_t A_t, \text{clip}(r_t, 1 - \varepsilon, 1 + \varepsilon) A_t)]$

$$\theta^{old} \rightarrow \theta^{new}$$

Source

Comparison



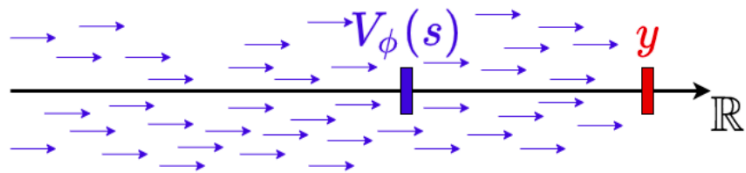
TRPO vs PPO

- + Very stable
 - + Theory (through NPG with Fisher info $\approx$ Surrogate with KL)
 - Works only for small models
 - Hard to implement
- $O(|\theta|^2)$
for iteration
- + Relatively easy to implement
 - + Works for big models (LLMs)
 - + Works better than TRPO
 - Many code-level optimisations

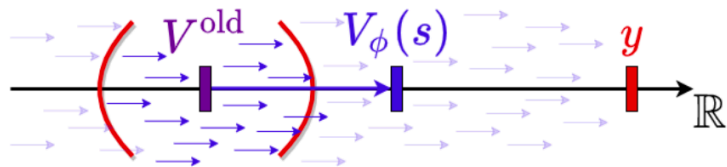
PPO Code-level Optimisations

- Value function clipping

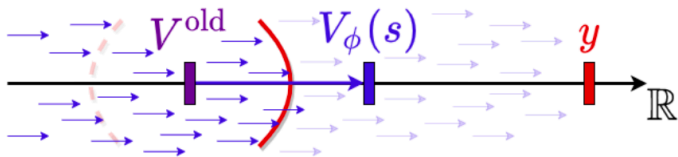
$$L^{Critic}(\theta) = \max[(V_\theta - y)^2, (\text{clip}(V_\theta - V_{\theta_{old}}, -\varepsilon, \varepsilon) - (y - V_{\theta_{old}}))^2]$$



$$(V_\theta - y)^2$$



$$\text{clip}(V_\theta - V_{\theta_{old}}, -\varepsilon, \varepsilon) - (y - V_{\theta_{old}})^2$$



$$L^{Critic}(\theta)$$

PPO Code-level Optimisations

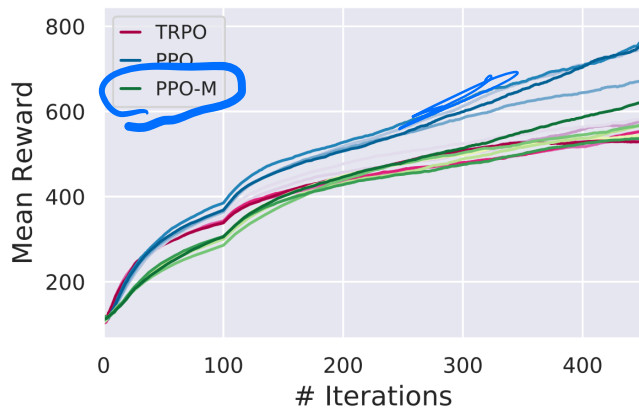
37

- Reward scaling (by standard deviation only)
- Reward clipping $[-10, 10]$
- Orthogonal initialisation and layer scaling
- Adam learning rate annealing
- Observation Normalisation
- Hyperbolic tan activation
- Global Gradient Clipping

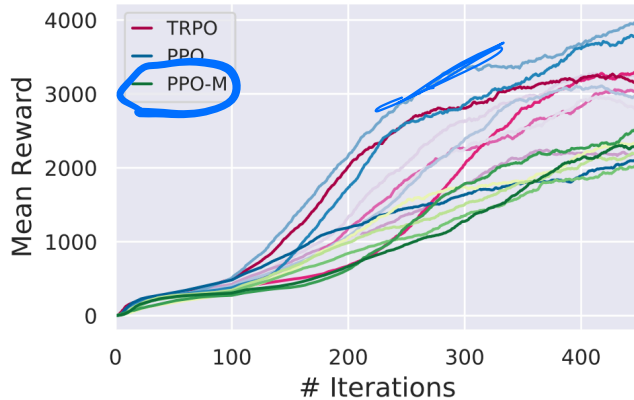
TRPO vs PPO

- + Very stable
- + Theory (approximately through NPG with Fisher info $\iff$ KL)
- Works only for small models
- Hard to implement
- + Relatively easy to implement
- + Works for big models (LLMs)
- + Works better than TRPO
- Many code-level optimisations

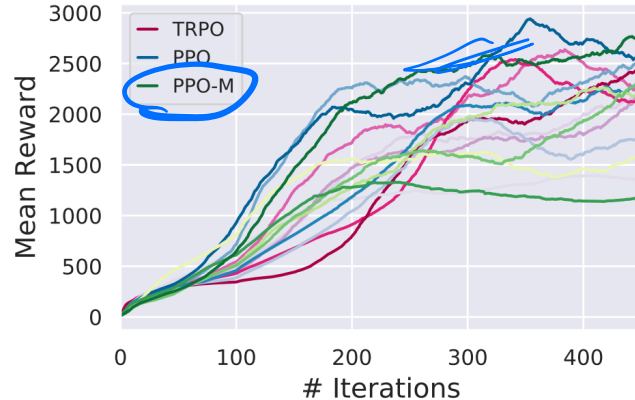
Humanoid-v2



Walker2d-v2



Hopper-v2



TRPO vs PPO

- + Very stable
- + Theory (through NPG with Fisher info $\approx$ Surrogate with KL)
- Works only for small models
- Hard to implement
- + Relatively easy to implement
- + Works for big models (LLMs)
- + Works better than TRPO
- Many code-level optimisations

TRPO / PPO vs REINFORCE / A2C?

- Optimization in π space instead of θ space (i.e., make steps in policy space directly)
- Can make more informative optimization step before collecting new samples (but usually need more samples in the batch...)
- Good example to see the difference: *gymnasium LunarLander*

Background

1. Practical RL course by YSDA, week 9
2. Reinforcement Learning Textbook (in Russian): 5.3
3. <https://spinningup.openai.com/en/latest/algorithms/trpo.html>
4. <https://spinningup.openai.com/en/latest/algorithms/ppo.html>
5. Implementation Matters in Deep RL
6. What Matters In On-Policy Reinforcement Learning?
7. 37 implementation details of PPO

Thank you for your attention!